Completeness of Relational Operations in Language Models

Results on a 500-Question Benchmark

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Abstract

We report results for five language models on a 500-question benchmark that asks for *every* item satisfying a condition, over records whose exact answers are obtained by executing SQL. On the 365 questions whose records are in the prompt, mean recall is 0.966–0.996, but only 81–98% of answers are exactly the correct set; for four of the five models nearly all errors are omitted items. The chance of a complete answer falls both as the number of correct items grows and as the table grows (pooled odds ratios of 0.50 and 0.39 per doubling, each adjusted for the other and for input format). Alternative prompts did not improve completeness, and models that stayed complete used far more reasoning tokens on larger answer sets.

1 Process

Benchmark. The benchmark has 500 questions over 575 synthetic tables (110,146 records) about fictional entities, so answers cannot come from memory. Each question is generated with the SQL query that defines it, and its gold answer is that query's result in SQLite. The 400 *in-context* questions put all records in the prompt; 60 *memory-only* questions filter fixed lists (U.S. states, chemical elements, presidents' surnames); 40 *retrieval* questions supply the top-k BM25 results. For in-context enumeration, exactly $g \in \{5, 10, 20, 50, 100\}$ of $n \in \{50, 200, 500, 1000\}$ rows match. Answer size and table size vary largely independently, but answers of 50 or more items never occur in 50-row tables (correlation of the log sizes 0.22), so trends are estimated with both sizes and the input format in one model. Records are shown as Markdown, CSV, JSON Lines or English sentences, rotating across questions. Table 1 lists the slices.

Models. DeepSeek V4 Pro and V4 Flash (OpenCode Go API), Claude Haiku 5.5 (Claude Code CLI 2.1.296, no tools) and MiMo-V2.6 Pro and Flash (OpenRouter), each with its default reasoning, temperature 0 where supported, and a 32,768-token output limit. Answers are returned as JSON.

Metrics. For gold set G and predicted set P: recall $|P \cap G|/|G|$, precision $|P \cap G|/|P|$, and **exact-set accuracy** $\mathbb{1}[P = G]$, the primary measure. Each error is classified as *missing*, *near-miss* (a garbled copy of a missing item), *wrong* (in the data but not qualifying), *made-up* (not in the data) or *duplicate*. An answer *omits* if it misses a correct item and *commits* if it adds an incorrect one. Answers that were truncated at the output limit, could not be parsed, or were refused by a safety filter are *invalid*: they count as incorrect for exact-set accuracy but are excluded from error-type counts, so a cut-off answer is not mistaken for omitted items.

Runs and statistics. **E1**: all 500 questions, direct prompt. **E2**: a stratified 100-question subset with a "be exhaustive" prompt and a "check every record first" prompt. **E3**: the subset again with the direct prompt. Intervals are 95% (Wilson for proportions, question-level bootstrap for means); model and prompt comparisons use paired question-level differences; trends with answer size use a logistic regression on $\log_2$ of the number of correct items.

Slice	Qs	Variants	Answer size	Context (k tok.)
Enumeration	100	one category filter	5–100	7.6
Enumeration (memory only)	60	U.S. states, elements, presidents	2–50	0.0
Enumeration (retrieval)	40	BM25 top-k, $k \in \{10, 20, 50\}$	5–40	1.1
Selection	60	1–3 conditions	1–92	4.1
Projection	40	DISTINCT; 2–3 attributes	7–39	2.9
JOIN	60	1:1, 1:N, M:N	3–55	3.4
Aggregation	50	COUNT, SUM, MIN/MAX/AVG, GROUP BY	3–367	2.8
Nested query	40	IN, NOT IN, correlated, scalar	3–32	2.7
Set operation	50	UNION, INTERSECT, EXCEPT, 3-way	4–97	0.7

Table 1: Benchmark composition. Answer size is the number of correct items (for aggregation, records aggregated); context is the median data size in the prompt.

2 Results

2.1 Overall completeness

Table 2 shows the 365 in-context set-valued questions. Recall and precision are close to 1 for every model, but exact-set accuracy ranges from 98% to 81%. The best model, DeepSeek V4 Flash, is statistically tied with DeepSeek V4 Pro; the other three are significantly below it, by 7 to 17 percentage points.

Model	Exact-set acc. (%)	Δ vs. best (pp)	Mean recall	Mean prec.	Omit. (%)	Comm. (%)	Invalid
DeepSeek V4 Pro	98 [96, 99]	−0.3 [-2.5, +1.6]	0.996 [0.991, 1.000]	1.000	2	0	1
DeepSeek V4 Flash	98 [96, 99]	(reference)	0.996 [0.989, 1.000]	0.999	1	1	1
Claude Haiku 5.5	91 [88, 93]	−7.1 [-10.1, -4.1]	0.995 [0.993, 0.997]	0.999	9	1	0
MiMo-V2.6 Pro	90 [86, 93]	−8.2 [-11.8, -4.9]	0.987 [0.978, 0.994]	0.997	7	4	2
MiMo-V2.6 Flash	81 [77, 85]	−16.7 [-20.8, -12.3]	0.966 [0.952, 0.979]	0.976	15	10	2

Table 2: In-context set-valued questions (E1, $n = 365$ per model), with 95% intervals. Δ: paired difference in exact-set accuracy from the best model. Mean recall counts invalid answers as 0; mean precision is over valid answers only. *Omit./Comm.*: share of answers that omit a correct item / add an incorrect or duplicate one. *Invalid*: truncated, unparseable or refused answers, counted as incorrect.

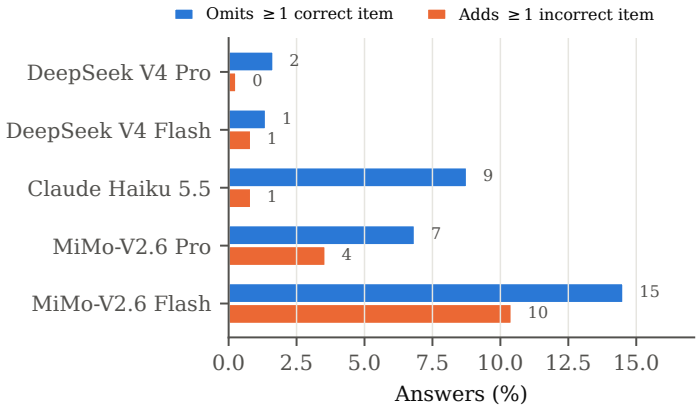


Figure 1: Share of in-context answers that omit at least one correct item, and that add at least one incorrect item (E1).

Errors are almost all omissions (Figure 1; counts in Table 12). Four models together returned 25 wrong items and 0 made-up items across all their answers. Incomplete answers usually miss only a few items (median 1; 64% miss at most a tenth of the answer), which is why recall stays high. MiMo-V2.6 Flash is the

exception: it returned 1,032 made-up items, 991 of them on a single 100-item question. Box 1 shows one example of each. Across all models, 7 of the 2,500 E1 answers were invalid, including 2 refused by a safety filter (Table 13).

Box 1. Two failures from E1

Omission (DeepSeek V4 Pro, `enum-ctx-029`). Prompt: 500 employee records as CSV (≈8k tokens), then *"List the names of all employees whose department is Support."* Gold answer: 50 names. The model returned 49 names, all correct, and omitted Kenneth Grimes (row 321 of 500). In three re-runs the answer was incomplete every time, omitting 2–7 names, a different set each time (Jose Greene in all three), and two of the three re-runs also added incorrect names.

Fabrication (MiMo-V2.6 Flash, `enum-ctx-053`). Prompt: 1,000 book records; *"List the titles of all books whose genre is Romance."* Gold answer: 100 titles. The model returned 1,125 distinct titles (plus 245 repeats): 66 correct, 68 real but not matching, and 991 that do not exist in the data, for example "Blooming Season", "Cold Garden", "Crimson Orchad", "Ember of Empty Phevi".

Takeaway: high recall does not mean complete answers; models leave correct items out far more often than they add wrong ones, so incompleteness is not visible in what they list.

2.2 Completeness, answer size and table size

On in-context enumeration (Figure 2, Table 3), every model is complete on at least 93% of questions with up to 10 correct items. With 50 or more, Claude Haiku 5.5 falls to 53%, MiMo-V2.6 Pro to 75% and MiMo-V2.6 Flash to 34%. Because large answer sets occur only in larger tables, Table 3 estimates both effects together, adjusted for input format. Pooled across models, each doubling of the answer set multiplies the odds of a complete answer by 0.50 (95% CI 0.36–0.60), and each doubling of the table by 0.39 (0.22–0.56): both factors matter on their own. Per model, the answer-size effect is significant for Claude Haiku 5.5 and MiMo-V2.6 Flash, and the table-size effect for Claude Haiku 5.5 and both MiMo models. The DeepSeek models fail too rarely (4 and 2 of 100 questions) for either effect to be estimated.

Takeaway: larger answers and larger tables each make a complete answer less likely.

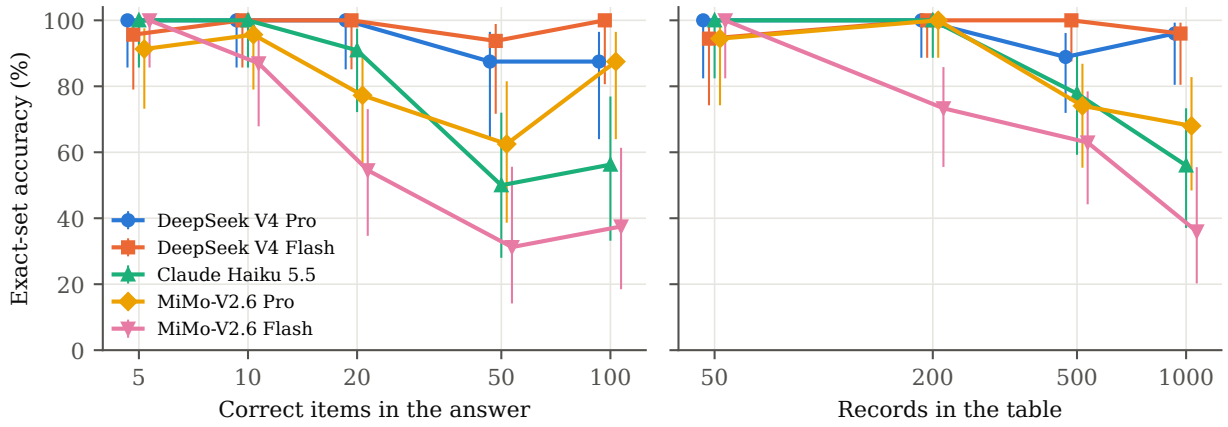


Figure 2: In-context enumeration (E1): exact-set accuracy by number of correct items (left) and table size (right), with 95% Wilson intervals.

Model	≤10 items (%)	≥50 items (%)	Failures	OR: answer size [95% CI]	OR: table size [95% CI]
DeepSeek V4 Pro	100	88	4	n.e.	n.e.
DeepSeek V4 Flash	98	97	2	n.e.	n.e.
Claude Haiku 5.5	100	53	17	0.17 [0.02, 0.28]	0.07 [0.01, 0.17]
MiMo-V2.6 Pro	93	75	16	0.79 [0.44, 1.20]	0.44 [0.11, 0.76]
MiMo-V2.6 Flash	93	34	34	0.33 [0.15, 0.44]	0.29 [0.08, 0.45]
All models (pooled)	97	69	73	0.50 [0.36, 0.60]	0.39 [0.22, 0.56]

Table 3: In-context enumeration (E1, 100 questions per model). Exact-set accuracy for small and large answer sets, and odds ratios of a complete answer per doubling of answer size and of table size, from one logistic regression that also adjusts for input format (below 1 means degradation). Pooled row: model fixed effects, bootstrap resampling whole questions. n.e.: not estimable (fewer than 5 failures).

2.3 By operation and input format

Every operation except set operations produced incomplete answers from at least three models (Table 4). Enumeration, which has the largest answer sets, is hardest; set operations are solved almost perfectly. Input format matters for some models (Table 5): MiMo-V2.6 Flash is complete on 73% of questions given as English sentences against 86% as JSON Lines, and Claude Haiku 5.5 on 88% of Markdown and 88% of sentence inputs against 96% for JSON Lines. The DeepSeek models are unaffected.

Takeaway: the effect is not specific to one operation; format effects are smaller and model-specific.

Operation	Qs	DeepSeek V4 Pro	DeepSeek V4 Flash	Claude Haiku 5.5	MiMo-V2.6 Pro	MiMo-V2.6 Flash
Enumeration	100	96	98	83	84	66
Selection	60	97	98	95	93	88
Projection	40	100	95	90	90	82
JOIN	60	100	98	93	88	82
Aggregation	15	93	93	93	87	67
Nested query	40	98	100	90	92	90
Set operation	50	100	100	100	98	100

Table 4: Exact-set accuracy (%) by operation, in-context set-valued questions (E1). Aggregation here is GROUP BY; scalar aggregates are in Table 7.

Input format	Qs	DeepSeek V4 Pro	DeepSeek V4 Flash	Claude Haiku 5.5	MiMo-V2.6 Pro	MiMo-V2.6 Flash
Markdown	105	98	98	88	90	87
CSV	97	96	97	93	93	78
JSON Lines	81	99	100	96	83	86
English sentences	82	99	98	88	94	73

Table 5: Exact-set accuracy (%) by input format, in-context set-valued questions (E1). Formats rotate across questions, so cells contain different questions.

2.4 Memory-only and retrieval settings

From memory, both DeepSeek models were always exact; the other three missed members of short fixed lists (Table 6), e.g. Claude Haiku 5.5 named Kentucky but not New Jersey as a state ending in "y". In the retrieval setting every model returned every correct record that was retrieved, so recall (0.869 for all models) equals the retriever's recall.

Takeaway: the retrieval slice, as built, measures the retriever rather than the models.

Model	Memory: exact (%)	Memory: recall	Retrieval: recall	Retrieval: recall over retrieved
DeepSeek V4 Pro	100	1.000	0.869	1.000
DeepSeek V4 Flash	100	1.000	0.869	1.000
Claude Haiku 5.5	87	0.952	0.869	1.000
MiMo-V2.6 Pro	83	0.955	0.869	1.000
MiMo-V2.6 Flash	83	0.958	0.869	1.000

Table 6: Memory-only and retrieval settings (E1).

2.5 Scalar aggregation

On 35 COUNT, SUM, MIN, MAX and AVG questions (Table 7), both DeepSeek models were always exact. Of the other models' 11 errors, 9 were undercounts.

Takeaway: wrong totals are mostly too low, consistent with dropped items.

Model	Qs	Exact (%)	Under-counts	Over-counts
DeepSeek V4 Pro	35	100	0	0
DeepSeek V4 Flash	35	100	0	0
Claude Haiku 5.5	35	94	2	0
MiMo-V2.6 Pro	35	91	3	0
MiMo-V2.6 Flash	35	83	4	2

Table 7: Scalar aggregation questions (E1).

2.6 Consistency across runs

Two independent samples of the subset agreed on 87–100% of verdicts (Table 8). We also re-ran 13 failures, chosen by hand to cover models and operations, three times each (Table 9). Some recurred every time: Claude Haiku 5.5 was incomplete on the same memory-only question (U.S. states containing "in") in all three re-runs, omitting Maine every time, and DeepSeek V4 Pro and MiMo-V2.6 Flash were incomplete on the same 50-item list in every re-run, though not always missing the same names (Box 1). In the two runs where MiMo-V2.6 Pro failed on that list, it omitted the same three employees. Others, including Claude Haiku 5.5's misses on selection, JOIN, projection and COUNT questions, never recurred.

Takeaway: some failures are systematic, but many are run-to-run variation, so completeness is best reported as a rate over many questions.

Model	Qs	Same verdict (%)	Questions that flipped
DeepSeek V4 Pro	87	98 [92, 99]	2
DeepSeek V4 Flash	93	100 [96, 100]	0
Claude Haiku 5.5	93	94 [87, 97]	6
MiMo-V2.6 Pro	93	91 [84, 96]	8
MiMo-V2.6 Flash	93	87 [79, 92]	12

Table 8: Agreement between two independent samples of the direct prompt (E1 vs. E3), with 95% Wilson intervals.

Model	Question	Answers	Records	Main-run outcome	Reproduced
Claude Haiku 5.5	`enum-par-013`	2	50	recall 0.50, 1 missing	2/3
Claude Haiku 5.5	`enum-par-021`	11	50	recall 0.73, 3 missing	3/3
Claude Haiku 5.5	`sel-029`	13	400	recall 0.85, 2 missing	0/3
Claude Haiku 5.5	`join-004`	55	100	recall 0.93, 4 missing	0/3
Claude Haiku 5.5	`nest-022`	13	200	recall 0.92, 1 missing	2/3
Claude Haiku 5.5	`proj-031`	37	300	recall 0.89, 4 missing	0/3
Claude Haiku 5.5	`agg-015`	12	400	answered 11 (gold 12)	0/3
Claude Haiku 5.5	`enum-ctx-029`	50	500	recall 0.96, 2 missing	0/3
DeepSeek V4 Pro	`enum-ctx-029`	50	500	recall 0.98, 1 missing	3/3
MiMo-V2.6 Flash	`enum-ctx-029`	50	500	recall 0.94, 3 missing, 3 wrong	2/3*
MiMo-V2.6 Pro	`enum-ctx-029`	50	500	recall 0.92, 4 missing	1/3
MiMo-V2.6 Flash	`enum-ctx-035`	100	1000	recall 0.52, 46 missing, 2 near-miss, 19 wrong, 371 made-up, 34 duplicate	2/3
DeepSeek V4 Flash	`enum-ctx-018`	5	50	recall 0.80, 1 near-miss	0/3

Table 9: Selected failures, each re-run three more times on the same model. *Reproduced*: re-runs again not exactly correct (* marks a re-run with no parseable answer, not counted). MiMo cases come from an earlier OpenCode Go run of the same models; DeepSeek and MiMo re-runs used OpenRouter.

2.7 Prompt variants

Neither the "be exhaustive" nor the "check every record" prompt changed exact-set accuracy significantly for any model; every paired interval includes zero (Table 10).

Takeaway: simple prompting does not fix incompleteness.

Model	Qs	Exhaustive − direct (pp)	Qs	Row-by-row − direct (pp)
DeepSeek V4 Pro	–	–	–	–
DeepSeek V4 Flash	93	−2.2 [-5.4, +0.0]	93	+0.0 [+0.0, +0.0]
Claude Haiku 5.5	93	+1.1 [-3.2, +5.4]	93	−2.2 [-6.5, +2.2]
MiMo-V2.6 Pro	93	+1.1 [-5.4, +6.5]	93	+4.3 [-1.1, +9.7]
MiMo-V2.6 Flash	93	+0.0 [-8.6, +8.6]	93	+6.5 [-2.2, +15.1]

Table 10: Paired change in exact-set accuracy (percentage points) from the direct prompt on the subset (E2), with 95% bootstrap intervals. DeepSeek V4 Pro's E2 runs are pending (provider usage limits).

2.8 Reasoning effort

Median reasoning length rose with answer size for the models that stayed complete (Figure 3): from 1,050 to 8,832 tokens for DeepSeek V4 Pro and from 173 to 6,442 for Claude Haiku 5.5. MiMo-V2.6 Flash often answered after one sentence of reasoning; its 61 answers with under 100 reasoning tokens were complete 62% of the time, against 72% for the rest.

Takeaway: completeness on large answer sets coincides with much longer reasoning; this is an association, not a controlled result.

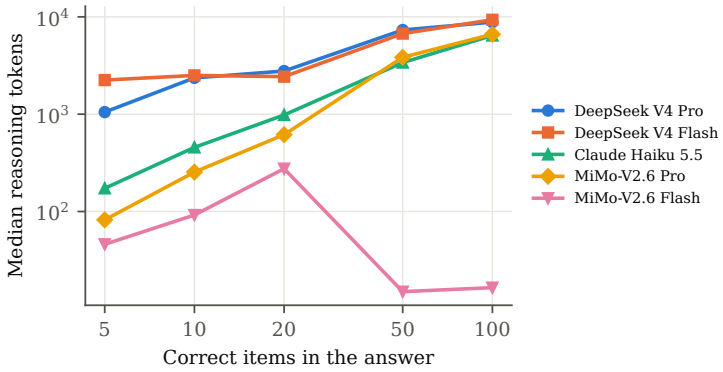


Figure 3: Median reasoning tokens on in-context enumeration by number of correct items (E1).

2.9 Usage

Across all runs the five models answered 3,796 calls; a further 1,163 attempts failed, almost all on provider rate or usage limits, and were retried. 77–97% of output tokens were reasoning (Table 11).

Model	Access	Calls	Failed	Input (M)	Output (M)	Reasoning (%)	Cost (USD)	Latency (s)
DeepSeek V4 Pro	OpenCode Go	594	565	3.83	2.21	97	6.87	23.4
DeepSeek V4 Flash	OpenCode Go	800	507	5.08	3.02	93	2.52	12.5
Claude Haiku 5.5	Claude Code CLI	802	91	8.04	1.88	87	2.49	7.7
MiMo-V2.6 Pro	OpenRouter	800	0	5.77	2.00	89	3.29	16.6
MiMo-V2.6 Flash	OpenRouter	800	0	5.78	1.72	77	1.25	13.1

Table 11: Usage across all experiments. *Calls*: model responses, including refusals; *Failed*: attempts that returned a provider error (mostly rate or usage limits) and were retried. †Runs of the same model through a second provider. Input and output in millions of tokens; *Reasoning*: share of output tokens; cost at provider list prices (Claude Haiku 5.5 and the OpenCode Go models were used through subscriptions); latency is the median per call.

3 Summary

- Mean recall of 0.966–0.996 corresponds to exact-set accuracy of only 81–98%; 1–15% of in-context answers omit at least one correct item.
- Errors are omissions, not additions, for four of five models; one model fabricates items on long lists.
- Completeness falls as answer sets grow and, independently, as tables grow (pooled odds ratios 0.50 and 0.39 per doubling).
- Prompt variants do not help; complete models reason far longer on large answer sets.
- The DeepSeek models are complete on 98% of in-context questions, so at this difficulty the benchmark leaves little headroom for the strongest models.
- Caveats: synthetic data from one seed; one sample per question in E1 (0–13% of verdicts change on a second sample); the answer-size trend rests on 100 enumeration questions per model; the retrieval slice measures the retriever; failure cases were chosen by hand; DeepSeek V4 Pro's E2 runs are pending (provider usage limits). A stopped Claude Haiku 4.5 run (90 questions) is excluded.

Code, data, raw responses and run manifests:
https://github.com/daryl-888/llm-relational-completeness.
Statistics follow E. Miller, "Adding Error Bars to Evals" (2024), https://arxiv.org/abs/2411.00640.

A Additional results

Model	Missing	Near-miss	Wrong	Made-up	Duplicate	Invalid answers
DeepSeek V4 Pro	6	0	1	0	0	1
DeepSeek V4 Flash	4	1	3	0	0	1
Claude Haiku 5.5	75	0	6	0	0	0
MiMo-V2.6 Pro	49	0	15	0	0	2
MiMo-V2.6 Flash	342	7	311	1,032	316	2

Table 12: Error counts by type on in-context set-valued questions (E1), counted on valid answers only; the last column is the number of invalid answers excluded.

Model	Qs	Truncated	Unparseable	Refused	API error
DeepSeek V4 Pro	500	1	0	0	0
DeepSeek V4 Flash	500	1	0	0	0
Claude Haiku 5.5	500	0	0	1	0
MiMo-V2.6 Pro	500	1	0	1	0
MiMo-V2.6 Flash	500	0	2	0	0

Table 13: Invalid answers among all 500 E1 questions: cut off at the 32,768-token output limit, not parseable as an answer, or refused by a safety filter (Claude Haiku 5.5: listing the elements with atomic numbers 55–86; MiMo-V2.6 Pro: one enumeration question stopped by the provider's content filter).

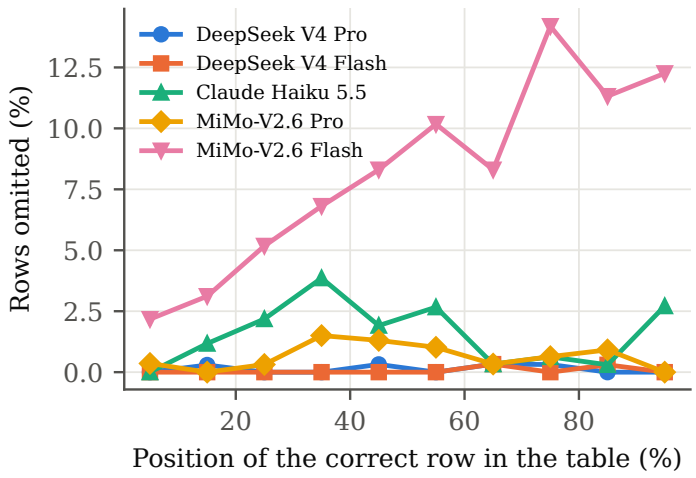


Figure 4: Share of correct rows omitted by the row's position in the table (in-context enumeration, E1). MiMo-V2.6 Flash omits rows near the end more often; the other models omit too few rows for a clear pattern.